FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Advanced)-2019

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. ABC

Sol. We know that

$I_{PQ} = I_x \cos^2\theta + I_y \sin^2\theta - I_{xy} \sin 2\theta$

$I_{PQ} = \frac{1}{4}mb^2\cos^2\theta + \frac{1}{4}ma^2\sin^2\theta \quad \left[\because I_{xy} = 0\right]$

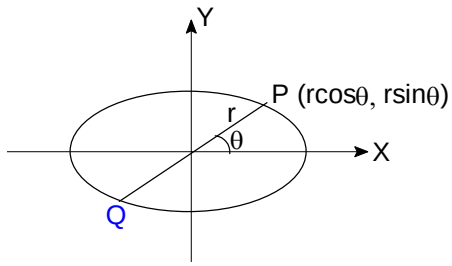
The equation of ellipse is

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{r^2}{a^2}\cos^2\theta + \frac{r^2}{b^2}\sin^2\theta = 1$

$a^2\sin^2\theta + b^2\cos^2\theta = \frac{a^2b^2}{r^2}$

$\therefore I_{PQ} = \frac{1}{4}m\left(b^2\cos^2\theta + a^2\sin^2\theta\right) = \frac{1}{4}m\frac{a^2b^2}{r^2}$



2. AD

Sol. $AB = AC = l$

$z = GN = OG\sin\theta$

$= \left(AO - AG\right)\sin\theta$

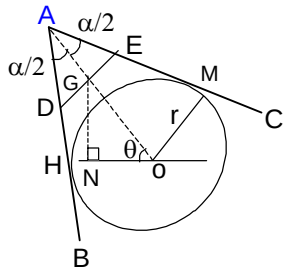
$= \left(r\cos ec\frac{\alpha}{2} - \frac{l}{2}\cos\frac{\alpha}{2}\right)\sin\theta$

$\frac{d\theta}{dt} = 0 \Rightarrow \cos\theta = 0 \; i.e; \; \theta = \frac{\pi}{2}$

$$\frac{d^2z}{d\theta^2} = -r\cos ec\frac{\alpha}{2} + \frac{l}{2}\cos\frac{\alpha}{2} \text{ for } \theta = \frac{\pi}{2}$$

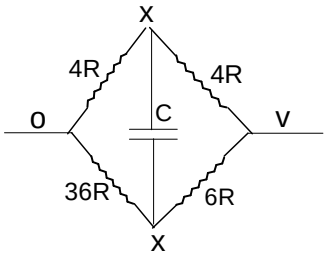
$$\text{Stable equilibrium}: \frac{d^2z}{d\theta^2} > 0 \Rightarrow l > 4r\cos ec\alpha$$

$$\text{Unstable equilibrium}: \frac{d^2z}{d\theta^2} < 0 \Rightarrow l < 4r\cos ec\alpha$$



3. BC

Sol. Applying KCL, we get



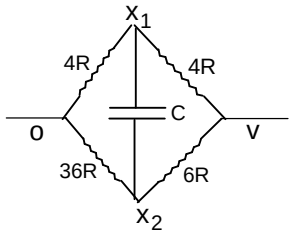
$$\frac{x-v}{4R} + \frac{x}{4R} + \frac{x}{36R} + \frac{x-v}{6R} = 0$$

$$x = \frac{3}{5}v$$

$$\therefore 4IR = \frac{2}{5}v$$

$$v = \frac{4\times 5IR}{2} = 10IR$$

In steady state:



$$\frac{x_1 - v}{4R} + \frac{x_1}{4R} = 0 \Rightarrow x_1 = \frac{v}{2}$$

$$\frac{x_2 - v}{6R} + \frac{x_2}{36R} = 0 \Rightarrow x_2 = \frac{6v}{7}$$

$$\text{change on capacities} = \frac{25}{7}CIR$$

4. ACD

5. AC

Sol. $T_1 = mg$ –(1)

$T_2 = mg - B$ –(1)

$$f = \frac{n_1}{2l}\sqrt{\frac{T_1}{\mu}} = \frac{n_1}{2l}\sqrt{\frac{T_2}{\mu}}$$

$$T_2 = \left(\frac{n_1}{n_2}\right)^2 T_1 = \left(\frac{n_1}{n_2}\right)^2 mg$$

$$B = mg - \left(\frac{n_1}{n_2}\right)^2 mg = mg\left[1 - \left(\frac{n_1}{n_2}\right)^2\right]$$

$$r = \left(\frac{3B}{4\pi\rho_\omega g}\right)^{1/3}$$

6. AC

Sol. $ab: \quad p_i v_i^{\gamma} = p_1 v_1^{\gamma} \quad -(i)$

$cd: \quad p_f v_f^{\gamma} = p_2 v_2^{\gamma} \quad -(ii)$

$bc: \quad p_1 v_1 = p_2 v_2 \quad -(iii)$

Dividing (2) by (1), we get

$$\frac{v_2}{v_2} = B^{\frac{2-\gamma}{\gamma-1}} = \sqrt{B}\left[\gamma = \frac{5}{3}\right]$$

work done by the gas

$$W = W_{ab} + W_{bc} + W_{cd}$$

$$= \frac{RT_0}{2}\ln B + \frac{3}{2}p_i v_i (1-B)$$

7. ACD

Sol. $\omega = \dfrac{qB}{m}$

The electron will not reach the anode only when it turns by $\frac{\pi}{2}$ by the time it comes near anode.

Let time taken be t.

$\theta = \omega t$

$$\frac{\pi}{2} = \frac{qB}{m}t \Rightarrow t = \frac{m\pi}{2qB}$$

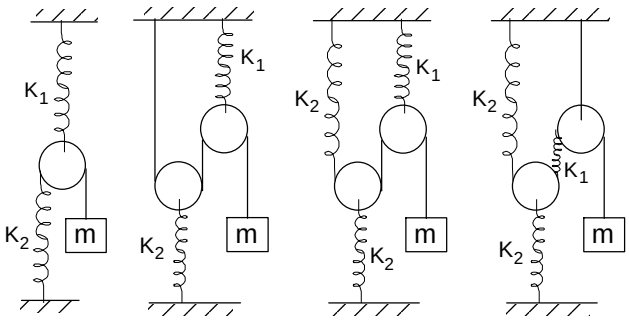
Acceleration due to electronic field $a = \dfrac{qE}{m}$

Horizontal distance curved in time t=d

$$\frac{1}{2}at^2 = d \Rightarrow \qquad B = \frac{\pi}{2}\sqrt{\frac{mE}{2qd}}$$

8. B

Sol.



$$\frac{1}{K_{eq}} = \frac{1}{K_2} + \frac{4}{K_1}$$

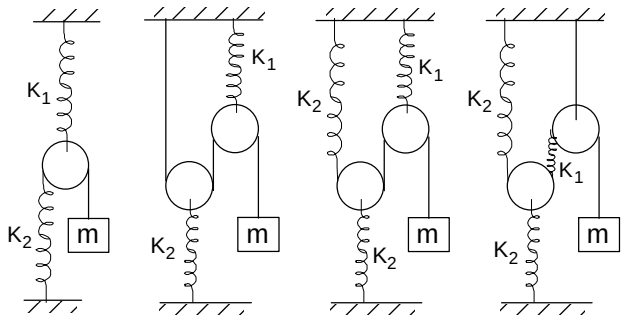
$$\frac{1}{K_{eq}} = \frac{4}{K_1} + \frac{4}{K_2}$$

$$\frac{1}{K_{eq}} = \frac{4}{K_1} + \frac{5}{K_2}$$

$$\frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{5}{K_2}$$

9. B

Sol.



$$\frac{1}{K_{eq}} = \frac{1}{K_2} + \frac{4}{K_1}$$

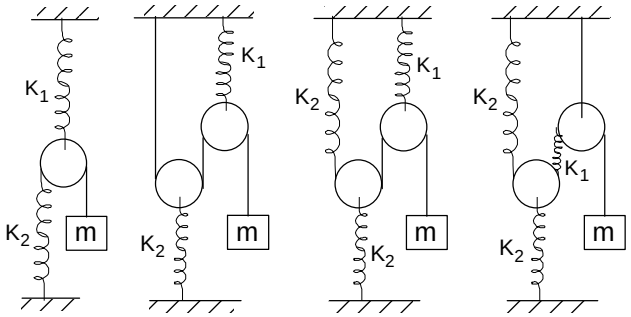
$$\frac{1}{K_{eq}} = \frac{4}{K_1} + \frac{4}{K_2}$$

$$\frac{1}{K_{eq}} = \frac{4}{K_1} + \frac{5}{K_2}$$

$$\frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{5}{K_2}$$

10. C

Sol.



$$\frac{1}{K_{eq}}=\frac{1}{K_2}+\frac{4}{K_1}$$

$$\frac{1}{K_{eq}}=\frac{4}{K_1}+\frac{4}{K_2}$$

$$\frac{1}{K_{eq}}=\frac{4}{K_1}+\frac{5}{K_2}$$

$$\frac{1}{K_{eq}}=\frac{1}{K_1}+\frac{5}{K_2}$$

11. C

Sol. For figure Column 1, Row1,

$$(\Delta x)_{at\ 0}=(\mu-1)\times 5.5\lambda=\left(\frac{52}{33}-1\right)5.5\lambda$$

Path difference $$=\frac{19}{33}\times 5.5\lambda=\frac{19}{6}\lambda$$

Hence, $$[n]=\left[\frac{19\lambda}{6}\times\frac{1}{\lambda}\right]=\frac{19}{6}=3$$

$$\delta=\frac{2\pi}{\lambda}(\Delta x)=\frac{19\pi}{3}$$

$$\therefore \textit{Intensity at } 0=4\mathrm{I}\cos^2\left(\frac{8}{2}\right)=3\mathrm{I}$$

For figure Column 1, Row2,

$$(\Delta x)_{at\ 0}=\left[(1.5-1)12\lambda\right]-\left[(1.2-1)15\lambda\right]=3\lambda$$

Hence, $$[n]=\left[3\lambda\times\frac{1}{\lambda}\right]=3$$

$$\delta=\frac{2\pi}{\lambda}(\Delta x)=6\pi$$

Hence, *Intensity at* $0=4\mathrm{I}$

For figure Column 1, Row3,

$$(\Delta x)_{at\ 0}=(\mu-1)D=\left(\frac{4}{3}-1\right)\times 10^7\lambda=\frac{10^7}{3}\lambda$$

Hence, $$[n]=\left[\frac{10^7}{3}\right]=33{,}33{,}333$$

$$\delta=\frac{2\pi}{\lambda}(\Delta x)=\frac{2\pi\times 10^7}{3}$$

Hence, *Intensity at* $0=\mathrm{I}$

For figure Column 1, Row4,

$$(\Delta x)_{at\ 0}=d\sin\theta\approx d\theta=\frac{8.82}{\pi}\times 10^4\lambda\times\left(\frac{1}{60}\times\frac{\pi}{180}\right)$$

$$=\frac{882}{108}\lambda=\frac{49}{6}\lambda$$

$$[n]=\left[\frac{49}{6}\right]=8,\quad \delta=\frac{2\pi}{\lambda}(\Delta x)=\frac{49\pi}{3}$$

Hence, *Intensity at* $0=3\mathrm{I}$

12. B

Sol. For figure Column 1, Row1,

$$(\Delta x)_{at\ 0} = (\mu - 1) \times 5.5\lambda = \left(\frac{52}{33} - 1\right) 5.5\lambda$$

Path difference $= \frac{19}{33} \times 5.5\lambda = \frac{19}{6}\lambda$

$$Hence, [n] = \left[\frac{19\lambda}{6} \times \frac{1}{\lambda}\right] = \frac{19}{6} = 3$$

$$\delta = \frac{2\pi}{\lambda}(\Delta x) = \frac{19\pi}{3}$$

$$\therefore Intensity\ at\ 0 = 4I\cos^2\left(\frac{8}{2}\right) = 3I$$

For figure Column 1, Row2,

$$(\Delta x)_{at\ 0} = \left[(1.5 - 1)12\lambda\right] - \left[(1.2 - 1)15\lambda\right] = 3\lambda$$

$$Hence, [n] = \left[3\lambda \times \frac{1}{\lambda}\right] = 3$$

$$\delta = \frac{2\pi}{\lambda}(\Delta x) = 6\pi$$

Hence, *Intensity at* $0 = 4I$

For figure Column 1, Row3,

$$(\Delta x)_{at\ 0} = (\mu - 1)D = \left(\frac{4}{3} - 1\right) \times 10^7\ \lambda = \frac{10^7}{3}\lambda$$

$$Hence, [n] = \left[\frac{10^7}{3}\right] = 33,33,333$$

$$\delta = \frac{2\pi}{\lambda}(\Delta x) = \frac{2\pi \times 10^7}{3}$$

Hence, *Intensity at* $0 = I$

For figure Column 1, Row4,

$$(\Delta x)_{at\ 0} = d\sin\theta \approx d\theta = \frac{8.82}{\pi} \times 10^4\ \lambda \times \left(\frac{1}{60} \times \frac{\pi}{180}\right)$$

$$= \frac{882}{108}\lambda = \frac{49}{6}\lambda$$

$$[n] = \left[\frac{49}{6}\right] = 8$$

$$\delta = \frac{2\pi}{\lambda}(\Delta x) = \frac{49\pi}{3}$$

Hence, *Intensity at* $0 = 3I$

13. D

Sol. For figure Column 1, Row1,

$$(\Delta x)_{at\ 0} = (\mu - 1) \times 5.5\lambda = \left(\frac{52}{33} - 1\right) 5.5\lambda$$

Path difference $= \frac{19}{33} \times 5.5\lambda = \frac{19}{6}\lambda$

Hence, $[n]=\left[\frac{19\lambda}{6}\times\frac{1}{\lambda}\right]=\frac{19}{6}=3$

$\delta=\frac{2\pi}{\lambda}(\Delta x)=\frac{19\pi}{3}$

$\therefore$ *Intensity at* $0=4\mathrm{I}\cos^2\left(\frac{8}{2}\right)=3\mathrm{I}$

For figure Column 1, Row2,

$(\Delta x)_{at\ 0}=\left[(1.5-1)12\lambda\right]-\left[(1.2-1)15\lambda\right]=3\lambda$

Hence, $[n]=\left[3\lambda\times\frac{1}{\lambda}\right]=3$

$\delta=\frac{2\pi}{\lambda}(\Delta x)=6\pi$

Hence, *Intensity at* $0=4\mathrm{I}$

For figure Column 1, Row3,

$(\Delta x)_{at\ 0}=(\mu-1)D=\left(\frac{4}{3}-1\right)\times10^7\lambda=\frac{10^7}{3}\lambda$

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$\delta=\frac{2\pi}{\lambda}(\Delta x)=\frac{2\pi\times10^7}{3}$

Hence, *Intensity at* $0=\mathrm{I}$

For figure Column 1, Row4,

$(\Delta x)_{at\ 0}=d\sin\theta\approx d\theta=\frac{8.82}{\pi}\times10^4\lambda\times\left(\frac{1}{60}\times\frac{\pi}{180}\right)$

$=\frac{882}{108}\lambda=\frac{49}{6}\lambda$

$[n]=\left[\frac{49}{6}\right]=8$

$\delta=\frac{2\pi}{\lambda}(\Delta x)=\frac{49\pi}{3}$

Hence, *Intensity at* $0=3\mathrm{I}$

SECTION – C

14. 5

Sol. $L_e=2\hbar$

$\frac{2}{5}mR^2\omega=2\hbar$

$\frac{2}{5}mRv=2\hbar$

$R=\frac{5\hbar}{mv}$

$R_{\min}=\frac{5\hbar}{mv_{\max}}=\frac{5\hbar}{mc}\qquad[n=5]$

15. 8

Sol. $R=\left(\frac{n-1}{n+1}\right)^2$

$\log R = 2\log(n-1) - 2\log(n+1)$

Differentiating

$\frac{dR}{R} = \frac{2dn}{n-1} - \frac{2dn}{n+1}$

$\frac{DR}{R} = 2\left[\frac{1}{n-1} - \frac{1}{n+1}\right]\Delta n = \left(\frac{4n}{n^2-1}\right)\frac{\Delta n}{n}$

$= \frac{8}{3} \times 3\% = 8\%$

16. 1

Sol. $Radiation\, pressure\ P = \frac{2E}{cA}$

$E = \frac{1}{2}PCA$

$= \pi m\omega \quad [n = 1]$

17. 0

Sol. $\phi = \oint \vec{E} \cdot \hat{n}\, ds = 4\pi q e^{-4r}$

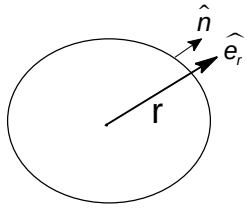
Change enclosed

$Q = \in_0 \oint \vec{E} \cdot \hat{n}\, ds$

$= 4\pi \in_0 qe^{-4r}$

Total change over all space $(r \to \infty)$

$Q = \lim_{r\to\infty} 4\pi \in_0 qe^{-4r} = 0$



18. 3

Sol. $\int_0^P dP = -\int_R^r \frac{4\pi G\rho^2}{3} r dr$

$= \frac{2\pi G\rho^2}{3}\left(R^2 - r^2\right) \qquad [n = 3]$

Chemistry

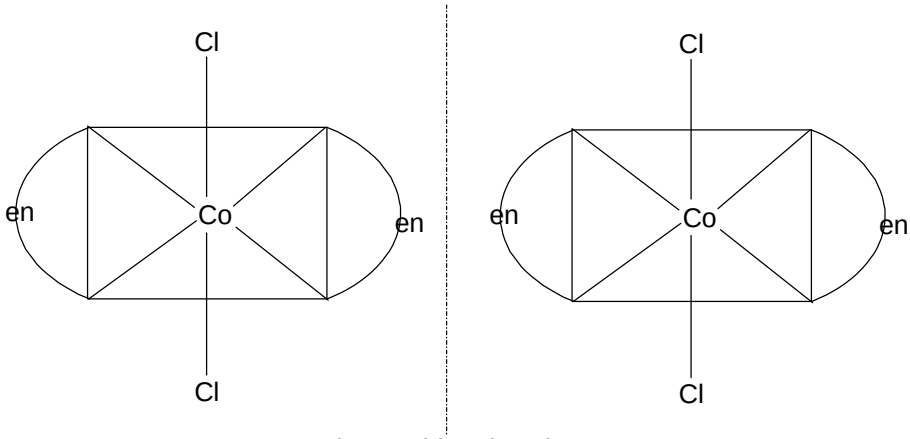
PART – II

SECTION – A

19. ABCD

20. ABC

Sol.



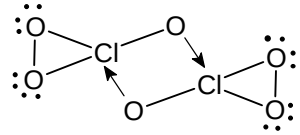
Superimposable mirror images

Optically inactive

21. ACD

22. AB

Sol. Cl_2O_6 in solid state $ClO_2^+ClO_4^-$

In the gaseous 

23. ABD

Sol. Fact

24. ABD

Sol. Salicylic acid is more acidic than benzoic acid due to o-effect.
o-toluidine is less basic than aniline due to SIP.

25. D

Sol. Dissolved $Al(OH)_3$ present in solution as $Al^{3+}(aq.)$ as well as $\left[Al(OH)_4\right]^-(aq.)$

$$\therefore S = \left[Al^{3+}(aq.)\right] + \left[Al(OH)_4^-(aq.)\right]$$

$$S = \frac{K_{sp}}{\left[OH^-\right]^3} + K_c\left[OH^-\right]$$

For minimum solubility

$$\frac{dS}{d\left(OH^-\right)} = 0$$

or $$-\frac{K_{sp} \times 3}{\left[OH^-\right]^4} + K_c = 0$$

$$\left[OH^-\right]=\left(\frac{3K_{sp}}{K_c}\right)^{\frac{1}{4}}$$

26. C
27. C
28. D

29. C
30. B
31. A

Sol. (for Q. 29 to Q. 31)

$P_4+3NaOH+3H_2O\longrightarrow PH_3(g)+3NaH_2PO_2$

$4FeS_2+11O_2\longrightarrow 2Fe_2O_3+8SO_2(g)$

$4H_3PO_3\xrightarrow{\Delta}3H_3PO_4+PH_3(g)$

$ClO_3^-+5Cl^-+6H^+\longrightarrow 3Cl_2(g)+3H_2O$

SECTION – C

32. 3

Sol. Mole of complex = $\frac{1}{266.5}$

1 mole of complex produce 2 mole of HCl

$\left[Cr(H_2O)_5Cl\right]Cl_2\longrightarrow\left[Cr(H_2O)_5Cl\right]^{+2}+2Cl^-$

Mole of HCl = $\frac{1}{266.5}\times 2$

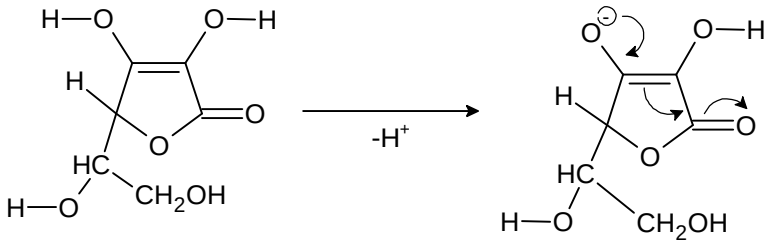
$N_{HCl}=\frac{2}{266.5\times 1}=7.5\times10^{-3}$

33. 7

Sol. $Z=\frac{PV}{nRT}=\frac{PM}{dRT}=\frac{8.314\times10^4\times18\times10^{-3}}{0.8\times8.314\times300}=\frac{18\times10}{300\times0.8}=\frac{3}{4}=0.75$

34. 2

Sol.



35. 6

Sol. $C_6(COOH)_6$ mellitic acid is produced

36. 0

Sol. $P_{ext}=0$

$\therefore$ Total work done $=P_{ext}(\Delta V)$

$\therefore P_{ext}=0$

$\therefore$ Work done $=0$

Mathematics

PART – III

SECTION – A

37. ACD

Sol. $f(x)=\dfrac{x^2-x+1}{x^2+x+1}$ *and* $g'(x)=6f'(x)\big(f(x)-1\big)\big(f(x)-2\big)$

Draw graph of f(x) and analyse.

38. AC

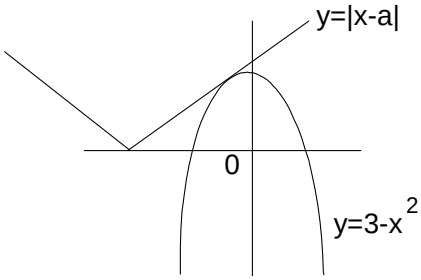
Sol. $f'(x)=\sin^{-1}(\sin x)\cdot 2\sin x\cos x+\cos^{-1}(\cos x)(-2\sin x\cos x)=0$

Hence, f(x) is a constant function.

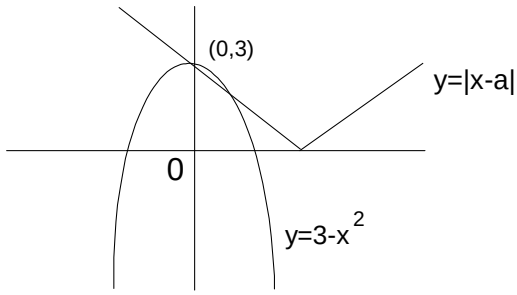
39. ABD

Sol. $x^2+|x-a|-3<0$

or $|x-a|<3-x^2$



Hence, $a=-\dfrac{13}{4}$



Hence, $a=3$

40. AB

Sol. $\vec{a}=-\hat{i}+2\hat{j}+\hat{k}$

Let $\vec{a}$ in new position be $\vec{b}=x\hat{i}+y\hat{j}+z\hat{k}$

$|\vec{a}|=|\vec{b}|\Rightarrow x^2+y^2+z^2=6$

$\vec{a}\cdot\vec{b}=0\Rightarrow -x+2y+z=0$

And $\vec{a},\vec{b}$ *and* $\hat{j}$ are coplanar

$$\Rightarrow\begin{vmatrix}-1 & 2 & 1\\ x & y & z\\ 0 & 1 & 0\end{vmatrix}=0 \qquad \therefore \vec{b}=\pm\sqrt{2}\left(\hat{i}+\hat{j}-\hat{k}\right)$$

41. B

Sol. $L_1: x=y=z$

$L_2: \dfrac{x-1}{1}=\dfrac{y+1}{-1}=\dfrac{z}{-1}$

Shortest distance=$1/\sqrt{2}$

Equation of plane containing L_2 and parallel to L_1 *is* $y-z+1=0$

42. ABCD

Sol. *Max* $\mathrm{P}(\mathrm{A}\cap\mathrm{B})=\mathrm{P}(\mathrm{A})=\dfrac{3}{5}$

$Min\, P(A \cap B) = P(A) + P(B) - 1 = \frac{4}{15}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{19}{15} - P(A \cap B)$

$P(A \cap \overline{B}) = P(A) - P(A \cap B) = \frac{3}{5} - P(A \cap B)$

$P\left(\frac{\overline{A}}{B}\right) = \frac{P(\overline{A} \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)} = 1 - \frac{3}{2}P(A \cap B)$

43. AB

Sol. $z_1 = a + ib = \cos\theta + i\sin\theta$

$z_2 = c + id = \cos\phi + i\sin\phi$

$\text{Re}(z_1\overline{z}_2) = 0 \Rightarrow \cos(\theta - \phi) = 0 \Rightarrow \theta - \phi = \pm\frac{\pi}{2}$

$|\omega_1| = \cos^2\theta + \cos^2\phi = 1$

Similarly, $|\omega_2| = 1$

$\text{Re}(\omega_1\overline{\omega}_2) = \cos\theta\sin\theta + \cos\phi\sin\phi$

$= \frac{1}{2}(\sin 2\theta + \sin 2\phi)$

$= \sin(\theta + \phi)\, cos(\theta - \phi) = 0$

Similarly, $\text{Re}(\overline{\omega}_1\omega_2) = 0$

44. B

45. C

46. D

Sol. (for Q. 44 to Q. 46)

In each of these questions, check options one by one.

47. A

48. B

49. D

Sol. (for Q. 47 to Q. 49)

$f(x) = \frac{7 + 6\tan x - \tan^2 x}{1 + \tan^2 x}; x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

or $y = \frac{7 + 6t - t^2}{1 + t^2}; t \in (-\infty, \infty)$

Hence, $y \in [-2, 8]$

$f(x) = 4\cos x + 3\cos\left(x + \frac{\pi}{2}\right) + 3;\ x \in R$

$= 4\cos x - 3\sin x + 3$

Hence, $f(x) \in [-2, 8]$

$f(x) = \sin^6 x + \cos^6 x;\ x \in R$

$= (\sin^2 x)^3 + (\cos^2 x)^3$

$= 1 - \frac{3}{4}(\sin 2x)$

Hence, $f(x) \in \left[\frac{1}{4}, 1\right]$

$f(x) = \cos x \cdot \cos\left(x + \frac{\pi}{3}\right) \cdot \cos\left(x - \frac{\pi}{3}\right); x \in R$

$= \frac{\cos 3x}{4}$

Hence, $f(x) \in \left[-\frac{1}{4}, \frac{1}{4}\right]$

SECTION – C

50. 7

Sol. Let C_0 be the circle passing through points A, B, C and D.

Radical axis of C_0 *and* $C_1 : x + 2y + 3 = 0$

Radical axis of C_0 *and* $C_2 : 2x + 3y + \lambda = 0$

Radical axis of C_1 *and* $C_2 : x - y - 3 = 0$

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & \lambda \\ 1 & -1 & -3 \end{vmatrix} = 0 \Rightarrow \lambda = 4$$

51. 0

Sol. $(M-I)^T = M^T - I$

$\Rightarrow (M-I)^T M = M^T M - M$

$\Rightarrow (M-I)^T M = I - M = -(M-I)$

$\Rightarrow \left|(M-I)^T M\right| = \left|-(M-I)\right|$

$\Rightarrow \left|(M-I)^T\right||M| = -|M-I|$

$\Rightarrow |M-I| \times 2 = -|M-I|$

$\Rightarrow |M-I| = 0$

52. 4

Sol. Coefficient of x^{70}

$= \left[(-8) + (-1 \times -7) + (-2 \times -6) + (-3 \times -5) + (-1 \times -2 \times -5) + (-1 \times -3 \times -4)\right]$

$= (-8 + 7 + 12 + 15 - 10 - 12) = 4$

53. 3

Sol. $z^9 = \frac{11 - 10iz}{11z + 10i}$ *Let* $z = a + ib\ (a, b \in R) : |z|^2 = a^2 + b^2$

$|z|^9 = \sqrt{\frac{11^2 + 220b + 10^2(a^2 + b^2)}{11^2(a^2 + b^2) + 220b + 10^2}}$

Clearly, $|z|^2 = a^2 + b^2 > 1$ *or* < 1 *is not possible.*

Hence $|z| = 1$

54. 1

Sol. $e^x - 1 - k\tan^{-1} x = 0$

or, $e^x - 1 = k\tan^{-1} x$

$y_1 = e^x - 1$ *and* $y_2 = k\tan^{-1} x$

Draw graphs and analyse.